ALICE L WALTON SCHOOL OF MEDICINE

Deep Vein Thrombosis

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Adapted from: Deep Vein Thrombosis

Listen to this Brick
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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of deep vein thrombosis (DVT).
2. Explain the pathophysiology of DVT.
3. Interpret clinical findings to properly diagnose DVT.
4. Outline the

CASE CONNECTION

PT is a 29-year-old female who comes to the emergency department with right leg pain. She tells you, "I thought it was just stiff, but it feels hot, and the back of my calf is sore to the touch." She has just returned from an extended trip abroad. "I got a great deal on the plane ticket, but taking two long flights trapped in a tiny seat was rough." She is generally healthy. Her only medication is oral contraceptive pills. On exam, her vital signs and O_2 saturation are normal. Her right calf is slightly erythematous, with tenderness that extends up into the popliteal fossa. The calf circumference is about 3 cm larger than on the unaffected side. The thigh is not tender. Her cardiopulmonary exam is normal.

What is the next best step in managing PT's condition? Consider your answers as you read, and we'll revisit PT at the end of the brick.

GO TO CONCLUSION ↓

What Is Deep Vein Thrombosis?

Deep vein thrombosis (DVT) refers to a blood clot that forms in the deep veins of an extremity, most typically in the legs.

A **thrombus** is a blood clot that forms in a blood vessel, most frequently in the venous system, although arterial thrombi can also occur. In contrast, an **embolus** is a piece of dislodged material or particulate matter that travels through the bloodstream from one place to another, where it may become lodged and cause obs urce of an embolus (i.e., either a part of, or the whole blood clot, dislodges from its site of origin and travels in the circulation to a new location), in which case the best terminology would be **thromboembolus** (Figure 1). But thrombi are not the only source for emboli, as an embolus can originate from other sources, too. Additional causes of embolization include tumors, foreign bodies, amniotic fluid, or even air.

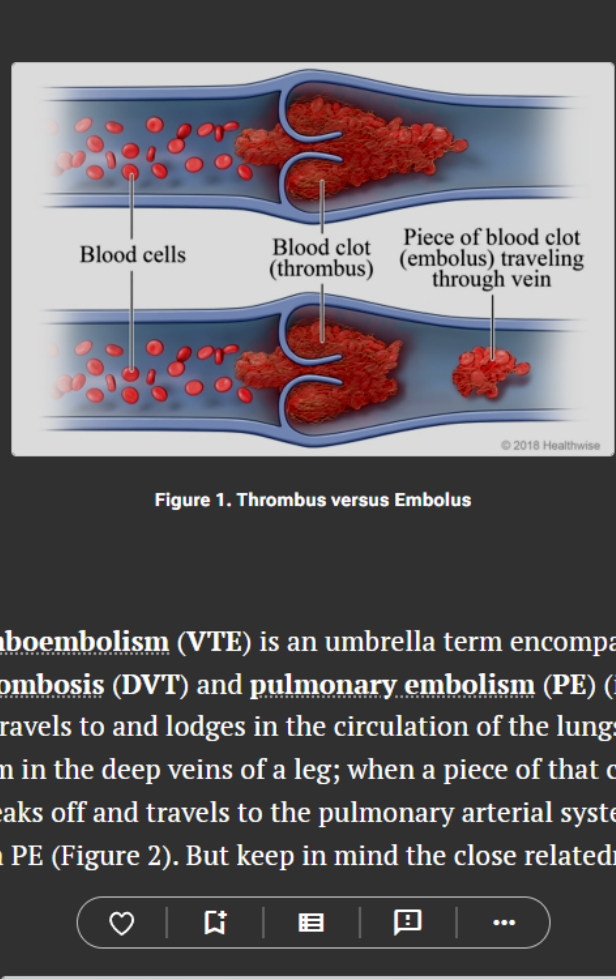
Blood cells
Blood clot (thrombus)
Piece of blood clot (embolus) traveling through vein
© 2018 Healthwise

Figure 1. Thrombus versus Embolus

Venous thromboembolism (VTE) is an umbrella term encompassing both **deep vein thrombosis** (DVT) and **pulmonary embolism** (PE) (i.e., an embolus that travels to and lodges in the circulation of the lungs). DVT most commonly form in the deep veins of a leg; when a piece of that clot (or the entire clot) breaks off and travels to the pulmonary arterial system, then it is referred to as a PE (Figure 2). But keep in mind the close relatedness of these terms.

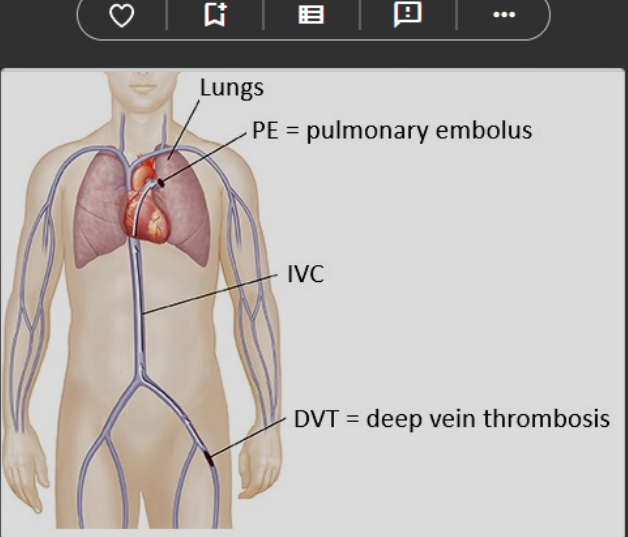

Figure 2. DVT and PE

What differentiates emboli from thrombi?

How Do Patients With Deep Vein Thrombosis Present? (LO #1)

Deep vein thrombi are found in the extremities, most commonly in the legs. The most common symptoms are **unilateral swelling of the leg**, often with **pain** and **redness** (Figure 3).

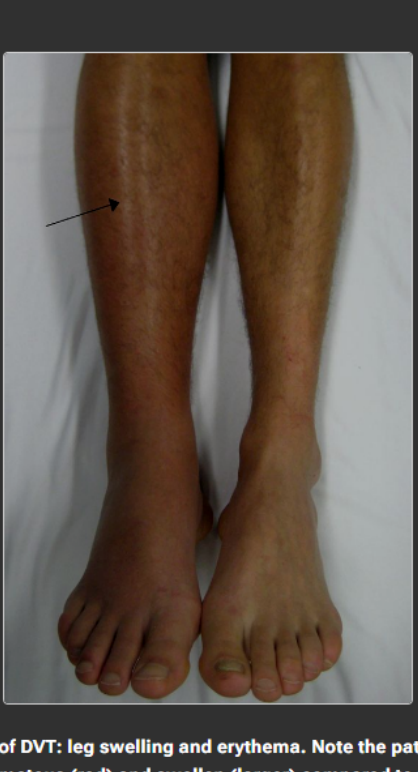

Figure 3. Clinical Presentation of DVT: leg swelling and erythema. Note the patient's right leg (black arrow) is both erythematous (red) and swollen (larger) compared to the left leg.

A palpable "cord" may be present—this occurs when the superficial veins in the involved leg are dilated and enlarged because of increased collateral circulation around the deep venous obstruction. In addition, the patient may experience increased pain with passive dorsiflexion of the ankle; this pain is known as **Homan sign** and is caused by irritation of the calf muscles.

Are symptoms of a DVT usually unilateral or bilateral?

What Is the Pathophysiology of Deep Vein Thrombosis? (LO #2)

The simplified goal of the coagulation system is to clot when injuries occur, and not to clot otherwise. However, pathologic states can disrupt this balance. There are three general conditions that can predispose to the formation of a thrombus, referred to as **Virchow's triad** (Figure 4). These conditions are:

1. Abnormal blood flow / stasis
2. Endothelial cell injury
3. Hypercoagulable states

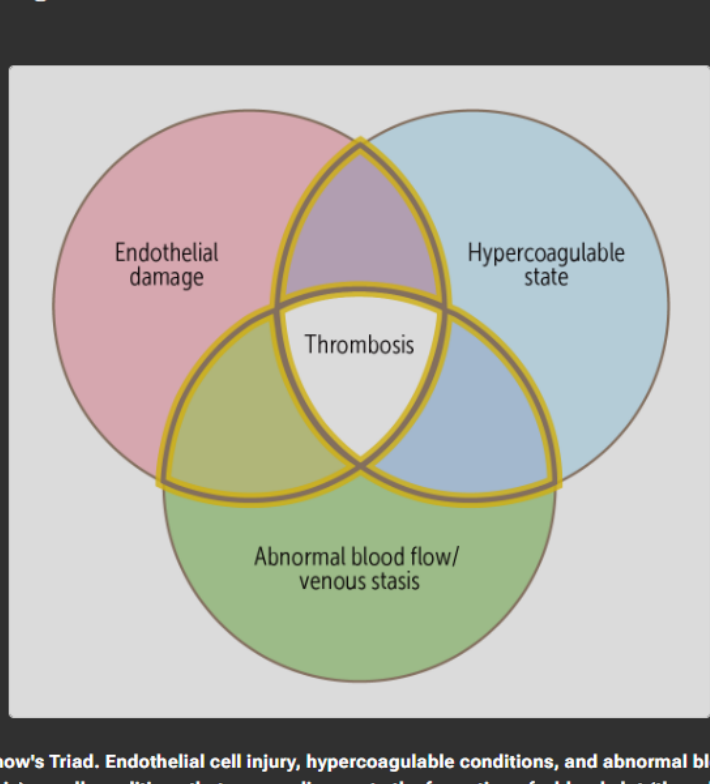

Figure 4 Virchow's Triad. Endothelial cell injury, hypercoagulable conditions, and abnormal blood flow (e.g., stasis) are all conditions that can predispose to the formation of a blood clot (thrombus).

What is the Virchow triad?

Risk Factors for Venous Thromboembolism

Let's review how these different factors can cause patients to develop clots:

- **Endothelial damage:** The coagulation cascade responds to any epithelial damage, whether from a cut from a knife (including surgery) or a ruptured atherosclerotic plaque. Any damaged endothelium can precipitate the formation of a clot. Some common causes of endothelial damage include smoking, hypertension, surgery, and trauma.
- **Abnormal blood flow:** Physiologically, blood is supposed to flow smoothly through the body, roughly parallel to the vessel wall (referred to as **laminar flow**). Laminar flow keeps platelets and clotting factors from interacting. However, in certain pathologic conditions, blood can either slow down (**stasis**) or move in a disorganized manner (**turbulence**) (Figure 5). Either change allows platelets and clotting factors to start mingling and potentially creating clots. Some common causes of abnormal blood flow are immobility (e.g., lying in a hospital bed or sitting during a long flight), central venous catheters, varicose veins, and atrial fibrillation.
- **Hypercoagulability:** Any medical condition that increases the risk of clot formation can inc s include cancer, use of estrogen-containing contraceptives, pregnancy, and obesity. Rarely, hypercoagulability may be due to hereditary thrombotic disorders like factor V Leiden mutation, prothrombin gene mutation, and protein C and S deficiencies. Hereditary causes should be suspected if a patient develops a DVT without any risk factors or precipitating events like immobility.

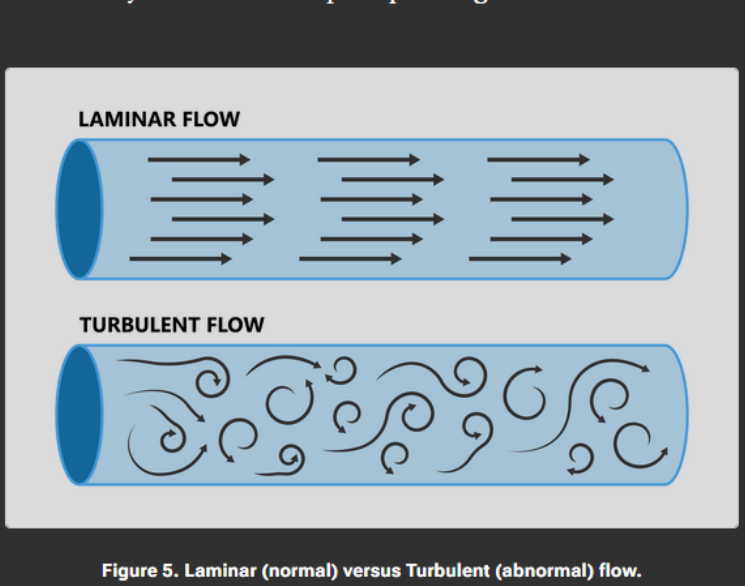

Figure 5. Laminar (normal) versus Turbulent (abnormal) flow.

CLINICAL CORRELATION

Factor V Leiden mutation is a mutation involving clotting factor V that renders the factor resistant to inactivation by activated protein C, which allows factor V to remain in an active state and predisposes the patient to abnormal thrombosis.

Factor V Leiden mut of hypercoagulability.

Complications

The most dreaded complication of DVT is **pulmonary embolism**, in which a clot lodges and obstructs a pulmonary vessel. This can be fatal, and it is why a diagnosis of DVT must be made accurately and timely.

Why are the lungs the major target of venous emboli? Because most DVT involve the legs, the common route for a thromboembolus is to travel up the inferior vena cava to the right side of the heart, through the right atrium and right ventricle, and into the pulmonary arterial circulation. Large clots may obstruct the large pulmonary arteries, whereas smaller clots may travel down to the level of the pulmonary arterioles or capillaries.

Less commonly, emboli can bypass the lungs. This occurs when an abnormal passage exists between the right and left heart (e.g., an atrial or ventricular septal defect). In that case, **"paradoxical emboli"** can pass directly from the right side of the heart to the left. From there, they can travel to the brain (leading to stroke) or to peripheral arteries supplying muscles or vital organs (leading to ischemic tissue injury).

Even after treatment, DVT patients may be at risk for development of **postthrombotic syndrome**, which refers to signs and symptoms of chronic venous insufficiency that develops in up to 50% of patients who have had a DVT. Postthrombotic syndrome causes leg pain, edema, skin ulcers, and discoloration, and is treated with leg elevation and compression.

Less commonly, pati (e.g., those with a massive iliofemoral DVT) may experience near complete occlusion of their venous drainage, leading to massive lower extremity edema, cyanosis, and/or gangrene. This condition is known as **phlegmasia cerulea dolens** (Figure 6). Unlike the other types of DVT, these patients receive urgent thrombectomy, usually surgically, since immediate removal of the clot is needed to preserve the limb or even the patient's life.

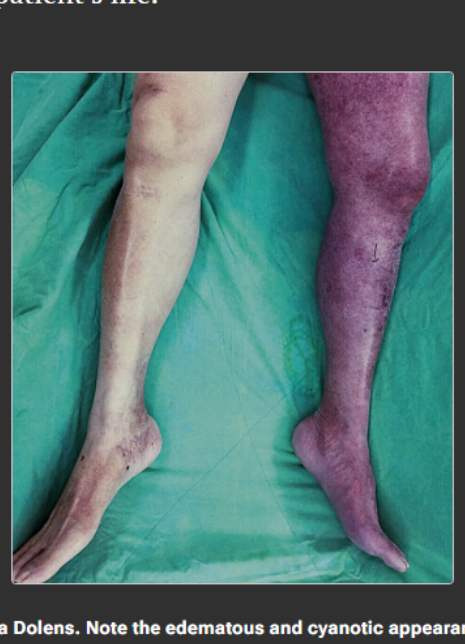

Figure 6. Phlegmasia Cerulea Dolens. Note the edematous and cyanotic appearance of the involved (patient's left) leg.

How Do We Diagnose Deep Vein Thrombosis? (LO #3)

Although clinical findings may point toward a diagnosis of DVT, a risk stratification tool is f dditional tests are necessary to confirm the diagnosis. These additional tests include D-dimer measurement and ultrasonography, as described below.

Risk Stratification

A clinical decision tool is used to answer the question: "What is the likelihood that this patient has a DVT?" This is called determining the **pretest probability** of the disease (i.e., the likelihood of the disease being present before any test is done). The **Wells score** for DVT is an example of such a tool (Table 1).

Table 1 Wells Score for Deep Vein Thrombosis

Clinical characteristics	Score
Active cancer (treatment within 6 months or palliative)	1
Paralysis, paresis, or recent cast immobilization of the lower extremities	1
Recently bedridden for ≥3 days, or major surgery requiring general or regional anesthesia within the previous 12 weeks	1
Localized tenderness along the deep venous system	1
Entire leg swollen	1
Calf swollen and ≥3 cm larger on symptomatic side than on asymptomatic side (measured 10 cm below tibial tuberosity)	1
Pitting edema in symptomatic leg	1
Collateral superficial veins	
Previously documented DVT	1
Alternative diagnosis at least as likely as DVT	−2

Patients with a low or moderate Wells score (e.g., 2 or below) will receive a D-dimer test to further evaluate the likelihood of DVT. Patients with a high Wells score (e.g., 3 or above) will typically start with ultrasonography.

D-Dimer. D-dimer is a fibrin degradation product elevated in the serum of patients who have a DVT. Physiologically, fibrinogen is constantly being converted to fibrin, resulting in detectable levels of D-dimer in all individuals, and these levels increase as we age. These levels may increase with certain pathologic conditions, to include thrombosis (Figure 7). Unfortunately, a D-dimer blood test is **sensitive** but not specific for DVT. This means that other conditions besides DVT can also cause an elevated D-dimer level, such as cigarette smoking, infection, surgery, and pregnancy. Interpreting an elevated D-dimer levels requires consideration of these other conditions.

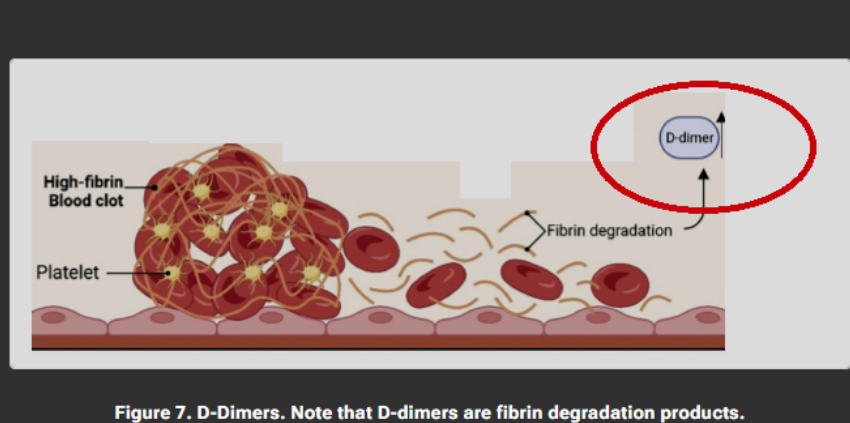

Figure 7. D-Dimers. Note that D-dimers are fibrin degradation products.

CLINICAL CORRELATION

The greatest utility of D-dimer testing in patients suspected of having a DVT is in ruling out the condition. Because the test is sensitive but not specific, then a patient with a normal (non-elevated) D-dimer level is highly unlikely to have a DVT.

Why is the D-dimer test effective in diagnosing DVT?

Ultrasonography. Compression ultrasonography is the best radiographic modality to determine if a patient has a DVT. This test is usually ordered in patients with moderate and high pretest probability for DVT as well as in those with a positive D-dimer. Under normal circumstances, pressure applied to a vein by the ultrasou mpletely collapse. Presence of a clot prevents the vessel from collapsing even when the clot may not be visualized (Figure 8).

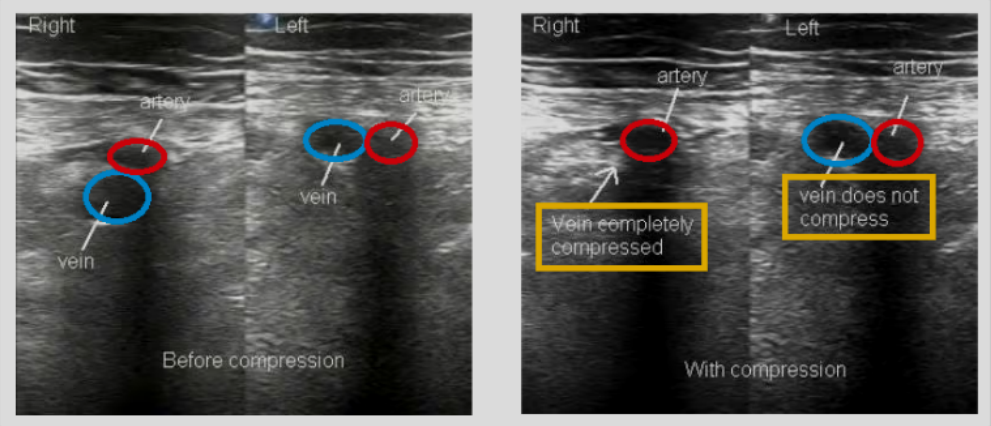

Figure 8. Compression Ultrasonography in the Diagnosis of DVT. Look at the before- and with-compression images of the patient's right leg. Note that in the before-compression image, the vein is visible, but once compression is applied, the vein collapses - this is normal. Now look at the patient's left leg. The vein is visible in both images - this is because the presence of a DVT keeps the vein open even with external compression. Thus, the correct interpretation of these images is a DVT involving a vein of the left leg.

Putting this together for an individual patient can be complex and critical, since the treatment is often a potent anticoagulant that has side effects. Treatment algorithms help clinicians make these choices (Figure 9).

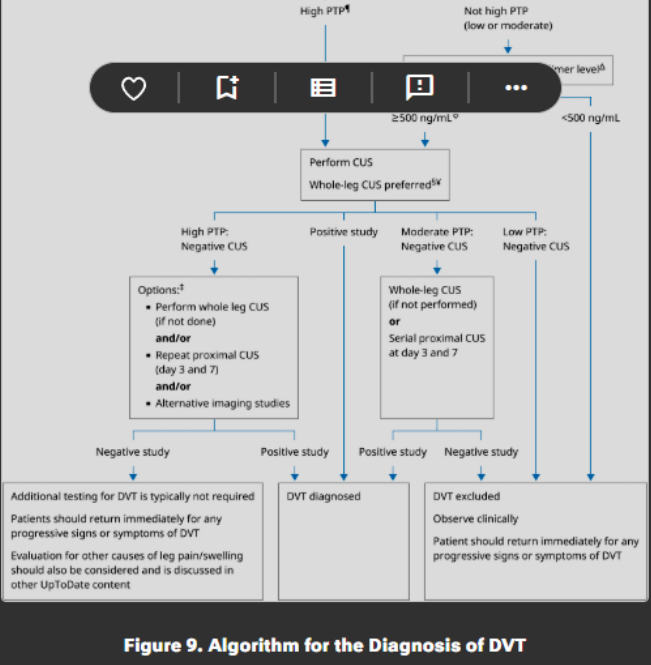

Figure 9. Algorithm for the Diagnosis of DVT

How Do We Manage Deep Vein Thrombosis? (LO #4)

DVT should be prevented in at-risk patients but then treated aggressively once diagnosed.

Prevention

Patients in hospitals are at higher risk of developing a DVT because of prolonged immobilization, trauma leading to endothelial damage, and/or infection and inflammation leading to hypercoagulability. To prevent this, patients at risk are often given anticoagulant agents while they are in the hospital as prophylaxis for DVT. This is most often **low-molecular-weight heparin** (e.g., enoxa (e.g., gastrointestinal bleeding, hemorrhagic stroke) who cannot take anticoagulants can be treated with mechanical leg-compression devices (Figure 10), but these are less effective than anticoagulants for preventing DVT.

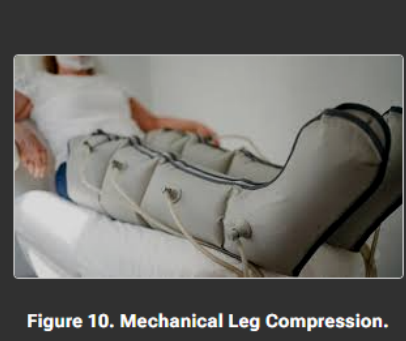

Figure 10. Mechanical Leg Compression.

Treatment

The treatment for DVT is **anticoagulation.** Anticoagulation prevents clots from forming or from getting bigger, allowing the body's anticoagulation system to break down the clot over time.

Traditionally, heparin is the immediate drug of choice for DVT, followed by warfarin for long-term anticoagulation. Over the past 20 years, low-molecular-weight heparin (enoxaparin) replaced intravenous heparin for the acute management of VTE. More recently, other oral anticoagulants, called **direct oral anticoagulants (DOACs)**, have become available, and some, such as rivaroxaban or apixaban, are now preferred for DVT treatment.

What is the main treatment used for DVT?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to PT, what is the next best step in managing her condition?

Your suspicion of DVT is confirmed by a lower extremity ultrasound. PT tells you that the ultrasound technician showed her some veins that would not flatten when compressed. You explain to PT how stasis of blood flow out of her legs during her flights is probably what allowed the clots to develop. You tell her that taking estrogen contraceptives is known to increase the risk of these clots. After PT's coagulation studies are checked and a pregnancy test is negative, she receives rivaroxaban, a direct oral anticoagulant, that she will continue taking for several months. You communicate with her primary care provider to facilitate close follow-up.

Summary

- Thrombi are clots that form in a vessel; deep vein thromboses (DVTs) are clots formed in the deep veins of an extremity.
- DVTs usually present as unilateral limb swelling, redness, and pain; they are most common in the legs but are possible in the upper limbs.
- Normal coagulation strikes a balance of clotting when appropriate but not to a pathologic extent.
- Patients with abnormal blood flow, endothelial dysfunction, and hypercoagulability are at risk for venous thromboembolic disease (Virchow triad).
- Complications of DVT include pulmonary embolus, postthrombotic syndrome (venous insufficiency), and phlegmasia cerulea dolens (occlusion of the iliofemoral veins).
- Although clinical findings may point toward a diagnosis of DVT, a risk stratification tool is used to determine what type of additional tests are needed for diagnosis.
- DVT is diagnosed by using D-dimer testing and deep vein duplex ultrasound.
- DVT is primarily treated by using anticoagulation.
- Direct oral anticoagulants, such as rivaroxaban or apixaban, are now preferred for DVT treatment.

Review Questions

1. A 19-year-old female patient presents to her college campus health care clinic complaining of pain and swelling in her right leg. She states that she has been sitting in the library for long hours for the past few days preparing for an organic che her leg last night and woke up this morning with the swelling. She is alert and oriented and denies shortness of breath, chest pain, and abdominal pain. A physical exam is unremarkable other than severe swelling and erythema in her left leg distal to the mid-thigh, with large palpable superficial veins. Her vital signs are normal. She currently uses a contraceptive vaginal ring for birth control and is allergic to codeine. Which of the following complications is most likely in this patient?

A. ○ Myocardial infarction
B. ○ Phlegmasia cerulea dolens
C. ● Pulmonary embolism
D. ○ Soft tissue infection
E. ○ Stroke

Hide Explanation

The correct answer is pulmonary embolism (C). This patient's leg pain, redness, and swelling, with a history of a large amount of time seated (stasis) over the past few days and use of estrogen contraceptives (hypercoagulability) make it likely that she has a deep vein thrombosis (DVT). This must be treated promptly to avoid pulmonary embolism, in which a clot dislodges from the DVT and lodges in the lungs with possible fatal consequences. Myocardial infarction (A) is due to arterial, not venous, occlusion. Phlegmasia cerulea dolens (B) is an uncommon occlusion of the iliofemoral arteries th tions (D), like cellulitis, are not common after DVT. Stroke (E) can occur as a rare complication of DVT via paradoxical emboli through congenital heart lesions like atrial or ventricular septal defects, but this patient lacks a murmur that would suggest this.

2. A 35-year-old female presents to the office for a swollen, hot, and painful left calf. She is taking an oral contraceptive but does not use any regular medication. She has never smoked. She drinks 5 units of alcohol a week. Her mother died of breast cancer at age 66 years. On examination, she has an erythematous, tender, warm left calf. Dorsiflexion of the foot produces pain at the calf, but the exam is otherwise normal. Which of the following most increased her risk for developing venous thromboembolic disease?

A. ○ Age
B. ○ Alcohol use
C. ● Estrogen contraceptive use
D. ○ Family history of breast cancer
E. ○ Smoking status

Hide Explanation

The correct answer is estrogen contraceptive use (C), which creates a hypercoagulable environment. This patient is 35 years old, so her age (A) does not increase the risk of v moderate alcohol use (B) does not increase her risk of VTE. Malignancy (eg, breast cancer) increases the risk of VTE. However, although the patient has a family history of breast cancer, she has no evidence of cancer at this time (D). Smoking is a major risk for VTE; however, the patient has never smoked, which decreases her risk and rules out smoking status (E).

3. A 45-year-old businessman sees his primary care doctor. He returned from a trip from China 2 days ago. He said he developed a "cramp" in his left calf during the flight and now has minor swelling in his left foot. He is concerned because his daily run caused discomfort in his calf. His vital signs are normal, and his exam is significant for minor pitting edema around his ankle. His left calf is very tender to deep palpation. Which diagnostic test should be ordered first?

A. ○ Complete blood count
B. ○ Compression ultrasonography
C. ○ CT scan with contrast of lower extremities
D. ● D-dimer assay
E. ○ No further workup is needed; follow up in 1 week

Hide Explanation

The correct answer i at moderate risk for DVT. If the D-dimer assay comes back low, then no further workup is necessary. A complete blood count (A) is unnecessary because there is little concern for infection, anemia, or thrombocytopenia. If the D-dimer is elevated, then compression ultrasonography (B) should be performed on the affected limb. A CT scan with contrast (C) is unnecessarily expensive and invasive when ultrasonography is an option. DVT should be treated as early as possible to improve outcomes, so no further workup is needed; follow-up in 1 week (E) is inappropriate.

4. What drug class is most likely used in the treatment of DVT?

A. ○ Anti-inflammatory
B. ○ Analgesics
C. ● Anticoagulants
D. ○ Antiplatelets

Hide Explanation

DVTs are treated with anticoagulants (D). The other listed drug classes - anti-inflammatory, analgesics, antiplatelets (A, B, D) - would not be considered as first-line or mainstay treatments for DVT.

How would you rate this Brick?

References

- https://www.aafp.org/afp/2012/1115/hi-res/afp20121115p913-f1.gif

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of deep vein thrombosis (DVT).~~
02 ☑ ~~Explain the pathophysiology of DVT.~~
03 ☑ ~~Interpret clinical findings to properly diagnose DVT.~~
04 ☑ ~~Outline the management plan of DVT.~~

Objective Confidence: 4/4

References

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